

RecipEZ: Architecture & Documentation

Michael Hannan | 3D Recipe Search Engine Pipeline

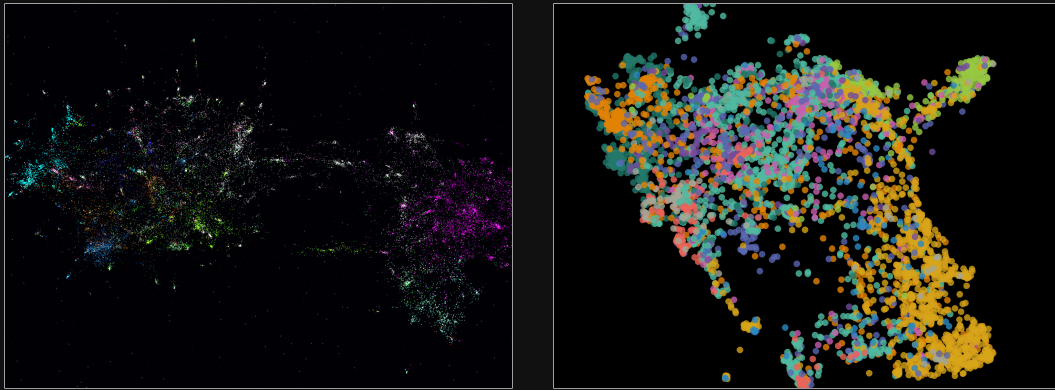


Figure 1: Shows comparison between V2.1 and V1.3

1. PREFACE

Wandering on the modern web is such a major nuisance. Scouring the web you can find bloat covered ads about practically any topic, everything is built to either gather data or make money off of you. RecipEZ was intended to be a fun personal project that helped solve this problem in the context of cooking and specifically recipes. Recipes tend to be a type of content that falls heavily for this bloat trap, with ads for cookware, meal kits, and grocery delivery services littering the search results and life stories about authors that have nothing to do with the actual recipe and that most people do not care about. The goal of RecipEZ was to create a simple, interactive search engine that accepts a list of raw ingredients and outputs the top recipe matches in a visually appealing 3D space.

2. BRAINSTORMING AND ARCHITECTURE RESEARCH

With the constraints defined—accepting a list of raw ingredients and outputting top recipe matches—several approaches were evaluated to establish a mathematical or semantic relationship between the input and the output.

2.1. Generative AI and Seq2Seq

The immediate thought was to simply pass the ingredients into a Large Language Model (LLM) and have it generate a recipe. However, this approach was discarded as it is computationally expensive, abstracts away the core technical challenge of building a search engine, and introduces hallucination risks. Fine-tuned Sequence-to-Sequence (Seq2Seq) models were also evaluated. While excellent for machine translation, they require massive amounts of training data and compute resources that were out of scope for this architecture.

2.2. Statistical Similarity

Jaccard Similarity, which measures the intersection divided by the union of two sets, was heavily considered. While useful for basic set matching, it fails to account for the frequency or relative importance of specific ingredients within a recipe. Furthermore, a core goal of RecipEZ was to create an interactive, visually appealing 3D space, and Jaccard metrics do not easily translate into the kind of distinct visual clusters required.

2.3. The Chosen Path: Clustering and Vectorization

The final architectural decision was to map recipes into a continuous vector space and cluster them based on culinary similarity. This required vectorizing the raw ingredient text into meaningful mathematical representations. To achieve this, the project relies on [word2vec](#).

3. THE DATA PIPELINE

The core engine of RecipEZ operates in three distinct phases to take text and turn it into a 3D searchable galaxy.

PIPELINE FLOW: INPUT → [word2vec](#) → [t-SNE](#) → [3D visualization](#)

3.1. Vectorization

The first step converts textual data—ingredients and instructions—into a numerical format processed by machine learning algorithms. [word2vec](#) weighs the importance of each word in the context of the entire dataset. Later iterations also explored SBERT, an advanced method using pre-trained language models to generate dense vectors that capture deeper semantic relationships.

3.2. Dimensionality Reduction

After vectorization, the data exists in an extremely high-dimensional space. To render this in a web browser, the dimensions must be reduced while preserving the local and global structure.

is used initially to discard noise and identify latent features. Following this, compresses the data into a 3D coordinate system, creating the clusters visualized in the final application.

3.3. Clustering and Searching

With the recipes mapped in 3D space, the actual search engine functionality is powered by . When a user inputs ingredients, the text is vectorized, placed into the space, and the algorithm retrieves the closest recipe nodes, which are then passed to the frontend.

4. DEVELOPMENT CHALLENGES AND SOLUTIONS

4.1. Data Loading and Parsing

A major hurdle during the initial stages was data normalization. Early datasets used highly inconsistent formatting (e.g., “3/4 cups”, “chopping onions”, “onions, chopped”). Because the pipeline treats unique text strings as separate features, these semantic equivalents were ruining the vector distance. After several attempts to write custom cleaning algorithms, the most effective solution was pivoting to an entirely separate, pre-normalized dataset to ensure clean vectorization.

4.2. Visual Tagging and Ontology

Converting words into 3D vectors results in visually meaningless dots on a screen, forcing the user to blindly interpret the clusters. To solve this, a “Tag Ontology” was developed. Utilizing metadata tags associated with the recipes (e.g., “vegan”, “dessert”), specific color codes were assigned to clusters. Plant-based tags were mapped to greens, while soups and stews were mapped to browns/beiges. This subjective but culturally intuitive color scheme made the space significantly more appealing and easier to navigate.

4.3. Web Visualization

Rendering this high-dimensional math in a browser required a robust frontend. was selected as the modern standard for web-based 3D graphics. While it featured a moderate learning curve, it was absolutely essential to achieving the interactable “galaxy” theme envisioned for the project.

4.4. Evaluating Accuracy

A lingering challenge is quantitatively testing the accuracy of the K-NN clusters. While visually the clusters make semantic sense, establishing a rigid metric to prove the search model is returning mathematically optimal recipes remains a primary focus for future iterations.

5. APPENDIX: TECHNICAL DEFINITIONS

TF-IDF (Term Frequency-Inverse Document Frequency)

An NLP technique that converts text into numerical vectors. It multiplies how often a word appears in a specific document (Term Frequency) by the inverse of how often it appears across all documents (Inverse Document Frequency). In RecipEZ, this naturally penalizes common ingredients like “water” or “salt”, while giving high mathematical weight to unique identifiers like “saffron” or “truffle oil”, allowing for distinct culinary clustering.

UMAP (Uniform Manifold Approximation and Projection)

A non-linear dimensionality reduction technique. Unlike linear methods (like PCA), UMAP excels at preserving both the local relationships (recipes right next to each other) and the global structure (how the dessert cluster relates to the meat cluster) when squashing high-dimensional TF-IDF vectors down into an interactable 3D visual space.

K-NN (K-Nearest Neighbors)

A supervised learning algorithm used here for spatial searching. Because TF-IDF creates a vectorized representation of the culinary universe, when a user inputs a query, K-NN simply measures the distance (often using Cosine Similarity) from the user’s input vector to the K closest recipe vectors in the database, returning them as the search results.

SVD (Single Value Decomposition)

A mathematical technique used to reduce data dimensionality by identifying latent features. In this pipeline, it acts as a preprocessing step before UMAP to filter out noise and improve performance. It ensures the 3D visualization captures the most important features of the recipes without affecting the underlying accuracy of the K-NN search.

Three.js

A cross-browser JavaScript library and API used to create and display animated 3D computer graphics in a web browser using WebGL. It provides the rendering engine required to actually draw the thousands of recipe data points as colored nodes in our “galaxy” visualization, handling camera controls, raycasting (for clicking recipes), and real-time rendering.

